Bernardo Rivas

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Department of Mathematics and Statistics
University of Toledo
Toledo, OH 43606, USA

Research Interests

Rigorous study of nonlinear dynamical systems using computational and combinatorial methods, with emphasis on Conley index theory and topological dynamics. Current directions include data-driven and hybrid dynamical systems, with applications to systems biology, robotics and control.

Employment

University of Toledo — Ohio, US
Postdoctoral Fellow, Department of Mathematics and Statistics — 2025 – Present
Supported by AFOSR MURI FA9550-23-1-0400 (PI: T. Lee, George Washington University; co-PI: W. Kalies), in collaboration with the University of Michigan

Education

Rutgers University — New Jersey, US
PhD in Mathematics, Advisor: Konstantin Mischaikow — 2019 – 2025

Universidade de São Paulo — São Paulo, Brazil
MS and BS in Mathematics, MS Advisor: Marcio Gameiro — 2012 – 2019

Publications

Topological Dynamics via Learned Hybrid Systems — 2026
B. Rivas, W. Kalies, K. Iwasaki, A. Bloch, M. Ghaffari
Proceedings of the 8th Annual Learning for Dynamics and Control Conference (L4DC), PMLR 331:1663–1674

Preprints

Computation of Attractor Lattices for a Class of Hybrid Systems — 2026
W. Kalies, **B. Rivas**, T. Wehbe
arXiv:2610.15123

Characterizing High-dimensional Dynamics by Combinatorial-Topological Methods on a Latent Space — 2026
P. Bailon, M. Gameiro, B. Gelb, W. Kalies, M. Kramar, K. Mischaikow, **B. Rivas**, E. Vieira
Submitted to *SIAM Journal on Applied Dynamical Systems*, arXiv:2609.01509

Conley Index Theory for Hybrid Systems — 2026
B. Rivas, W. Kalies
In revision, *Nonlinear Analysis: Hybrid Systems*, arXiv:2606.18501

Boolean Models Coarsely Sample Continuous Dynamics of Regulatory Networks — 2026
B. Cummins, M. Gameiro, T. Gedeon, K. Mischaikow, **B. Rivas**
In revision, *PLOS Computational Biology*, arXiv:2606.14925

Global Dynamics of Ordinary Differential Equations: Wall Labelings, Conley Complexes, and Ramp Systems — 2024
M. Gameiro, T. Gedeon, H. Kokubu, K. Mischaikow, H. Oka, **B. Rivas**, E. Vieira
arXiv:2412.11078

In Preparation

Learning Cycling Signatures of Hybrid Systems
D. Hien, K. Iwasaki, **B. Rivas**

Topological Validation of Parameter Inference in Gene Regulatory Networks
W. El-Khateeb, W. Kalies, **B. Rivas**

Towards an Algorithmic Approach to Control of Dynamic Phenotypes
M. Gameiro, W. Kalies, K. Mischaikow, **B. Rivas**

Talks

Characterizing global dynamics of hybrid systems — New Brunswick, NJ
Applied Math Seminar, Rutgers University — 2026

Homological Dynamics — New Brunswick, NJ
Conference on Algebra, K-theory and Applications — 2026

Computational Dynamics — Toledo, OH
UToledo Mathematics Colloquium — 2026

Computational Conley Theory — Vitória, Brazil
UFES – Universidade Federal do Espírito Santo — 2026

A computational topology approach to gene regulation — Austin, TX
Texas-Louisiana Section of SIAM 8th Annual Meeting — 2025

Global dynamics in systems biology — Denver, CO
SIAM Dynamical Systems — 2025

Computational dynamics of gene regulatory networks — Athens, OH
Dynamics Seminar — 2025

Network dynamics via combinatorics, topology and homology — Toledo, OH
AMS Seminar — 2025

Analytical bounds for combinatorial topological dynamics — Montreal, Canada
Workshop – Combinatorial topological framework for nonlinear dynamics — 2024

A computational framework for dynamics of genetic regulatory networks — Washington, DC, US
AMS Spring Eastern Sectional Meeting — 2024

Combinatorial topological method for gene regulatory networks — Leiden, Netherlands
Workshop – Computing combinatorial dynamics in high dim. biological networks — 2023

Rigorous numerics of invariant manifolds — São Carlos, Brazil
Summer Meeting in Differential Equations — 2019

Mentoring

Informal research mentoring (not thesis advisor of record):

Wael El-Khateeb (now Assistant Professor, Emirates Aviation University) — University of Toledo
Topological validation of parameter inference in gene regulatory networks. — 2025 – 2026

Tony Wehbe — University of Toledo
Computational Conley theory for hybrid systems. — 2025 – Present

Software

hybrid_boxmap: Morse graphs, Conley indices and attractor lattices of hybrid systems, built on CMGDB — Python

MorseGraph-L4DC: Convex optimization and topological analysis for hybrid systems — Python

LatentDynamics: Topological analysis of learned dynamics on latent spaces — Python

DSGRN: Dynamic Signatures Generated by Regulatory Networks (core algorithms) — C++/Python

CMGDB: Conley-Morse Graph Database — C++/Python

Teaching

Course Development

DATA 5330 – Data Analytics and Visualization — University of Toledo — 2026 – 2027
Online M.S. in Data Science and AI

Instructor of Record

DATA4980 – Advanced Topics in Data Science	University of Toledo	Spring 2026
640:311 – Introduction to Real Analysis I	Rutgers University	Summer 2024
640:300 – Introduction to Mathematical Reasoning	Rutgers University	Summer 2023
640:252 – Elementary Differential Equations (online)	Rutgers University	Summer 2022
Minicourse on Differential Equations	ICMC-USP	Winter 2019

Teaching Assistant (Rutgers University)

640:311 – Introduction to Real Analysis I	Spring 2024
640:311 – Introduction to Real Analysis I	Fall 2023
640:244 – Differential Equations for Engineering and Physics	Spring 2023
640:481 – Mathematical Theory of Statistics	Fall 2022
640:421 – Advanced Calculus for Engineering	Fall 2022
640:244 – Differential Equations for Engineering and Physics	Spring 2022
640:135 – Calculus I for the Life and Social Sciences	Fall 2021

Poster Presentations

L4DC: Learning for Dynamics and Control, Los Angeles, CA, USA	2026
CBMS: Applied Mathematics and Machine Learning, Houston, TX, USA	2025
Workshop of Thesis and Dissertations, ICMC-USP, Brazil	2018
Algebraic Topology in Data and Dynamics, Montana State University, USA	2018

Conference & Workshop Participation

SIAM Annual Meeting (AN26), Cleveland, OH, USA	2026
Modularity of Biological Systems Workshop, NITMB, Chicago, IL, USA	2026
Id. of Gene Regulatory Network Function from Data, Providence, RI, USA	2022
International Congress in Mathematics, Rio de Janeiro, Brazil	2018
Summer Meeting in Differential Equations, ICMC-USP, Brazil	2017
GeoGebra Global Gathering, Linz, Austria	2015

Service

Minisymposium co-organizer, "Data-driven Topological Dynamics" (with W. Kalies), SIAM Conference on Mathematics of Data Science (MDS26)	2026
Referee, *Nonlinear Analysis: Hybrid Systems*	2026
Mentor, Directed Reading Program, Rutgers University	2024
Member, Graduate Student Liaison Committee, Rutgers University	2022 – 2025
Head Proctor, Rutgers University	2021 – 2023
Undergraduate representative, Mathematics Department Committee, USP	2015 – 2016

Fellowships

Research Internship Abroad, FAPESP – Rutgers University	2018
Master's Fellowship, FAPESP – Universidade de São Paulo	2017 – 2019
Science Without Borders, CNPq – Queen Mary University of London	2014 – 2015
Scientific Initiation Fellowship, FAPESP – Universidade de São Paulo	2013 – 2016